

IRIS-CAND-22

One-Command Pilot Orchestration

Timebox	5 hours	Project basis	IRIS-061
Primary stack	Python + PostgreSQL/PostGIS	Submission	Git repository / zip + run instructions

Objective

Create a lean one-command run that promotes accepted data, refreshes candidate views, and exports dossiers.

Scenario

The launch does not need a full orchestration platform. It needs a repeatable run with visible stale/failed stages.

Technical task

- Implement a Python CLI or Make target that executes ordered stages.
- Check prerequisites and stop/mark failed stages clearly.
- Refresh BESS and peatland materialized views.
- Export sample dossiers.
- Write concise run status with counts, timestamps, and failure details.

Required deliverables

- CLI
- Stage runner
- Run manifest
- Tests for success and one failing stage

Acceptance criteria

1. One command produces both vertical outputs.
2. A failed critical stage cannot be reported as success.
3. Stale view/export state is visible.
4. Rerun is safe.

Engineering constraints

- Use Python 3.12+ and PostgreSQL 16+ with PostGIS 3.4+ unless the assignment explicitly narrows the stack.
- Use a canonical geometry column named **geom**. Never silently invent missing geospatial attributes.
- Every persisted business entity must carry a non-null **country_code**; identifiers and joins must be country-scoped.
- Treat data completeness, CRS, units, source date, and uncertainty as explicit data contracts.
- Prefer reproducible CLI commands and SQL migrations over notebook-only work.
- No external production credentials are required. Use local fixtures/mocks where a source would otherwise be remote.
- Your submission must run locally from a clean checkout using the documented commands.

Evaluation rubric

Criterion	Weight
Orchestration correctness	30%
Failure visibility	25%
Idempotency	20%
Tests	15%
Simplicity	10%

Submission expectations

- README: setup, assumptions, commands, and architecture choices.
- Automated tests covering the highest-risk correctness conditions.
- A small deterministic fixture set committed with the submission.
- No production secrets, paid APIs, or inaccessible proprietary data.
- Document any deliberate simplification and how it would evolve for production.

Project context

This exercise is derived from the Project IRIS delivery playbook. The pilot prioritizes a one-region vertical slice, country-aware reusable adapters, PostgreSQL/PostGIS, geometry correctness, controlled promotion, explainable spatial screening, and two-page prospecting outputs. The challenge intentionally tests production-minded engineering without requiring access to private IRIS infrastructure.

Reference uncertainty wording used by the project:

Preliminary prospecting material. Figures, eco-point estimates and site suitability are indicative and based on available source data and commercial screening assumptions. The 8 eco-points/m2 factor is the current commercial baseline, not certified compensation. Ownership, planning, grid capacity, environmental eligibility and transferability remain subject to project-specific verification. No permit, reservation or construction readiness is represented.